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AI BLUEPRINT

Unified Omnichannel Customer Experience Platform

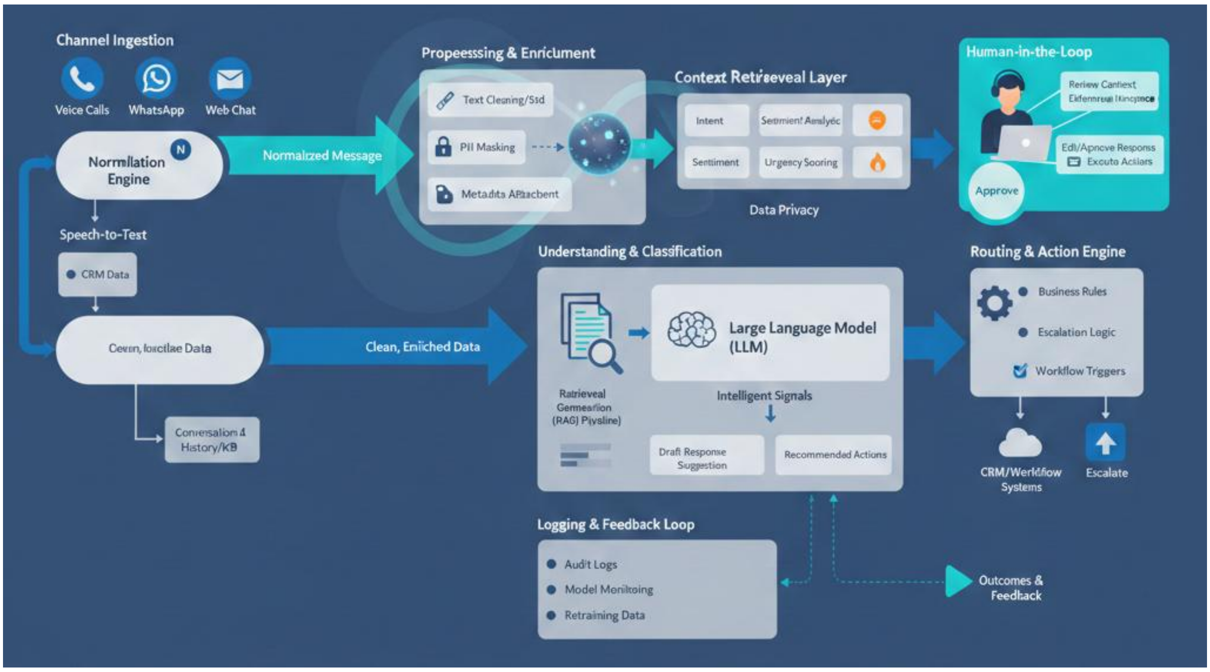

Image: Data flow of the whole user journey (AI generated image, may have some spelling errors)

1.Textual Data Flow (End-to-End)

Step 1: Channel Ingestion Layer

Customer interactions enter the platform from multiple sources:

- Voice calls (via telephony systems)
- WhatsApp
- Email
- Web chat
- Social platforms

Each input is converted into a **normalized message format**.
For voice calls, speech-to-text is applied before further processing.

Step 2: Preprocessing and Enrichment

Before any intelligence is applied:

- Text is cleaned and standardized
- Language detection is performed
- Personally Identifiable Information (PII) is masked or tokenized
- Metadata such as channel, time, and customer ID is attached

This ensures downstream models operate on safe and consistent data.

Step 3: Context Retrieval Layer

The system fetches relevant context, including:

- Past conversation history across all channels
- CRM data such as customer profile, tier, and previous tickets
- Knowledge base or policy documents

This context is stored temporarily in a **conversation state** for real-time use.

Step 4: Understanding and Classification

Multiple lightweight models operate in parallel:

- Intent classification determines what the customer is trying to do
- Sentiment analysis estimates emotional tone
- Urgency scoring predicts priority based on content, sentiment, and customer attributes

Outputs from these models are **structured signals, not free text.**

Step 5: Reasoning and Response Generation

For agent assistance:

- Relevant context is retrieved using a Retrieval-Augmented Generation (RAG) pipeline
- The language model generates response suggestions grounded in enterprise data
- The system also generates recommended next actions such as escalation or follow-up

No response is sent directly to the customer without agent review.

Step 6: Routing and Action Engine

Based on urgency, intent, and business rules:

- Conversations may be escalated
- Tickets may be created or updated
- Tasks may be triggered in CRM or workflow systems

This layer combines model outputs with deterministic rules.

Step 7: Human-in-the-Loop

Agents:

- Review conversation context
- Edit or approve suggested responses
- Execute recommended actions

All final customer-facing communication is human-approved.

Step 8: Logging and Feedback Loop

- Model outputs, agent actions, and outcomes are logged
- Data is stored for analytics, monitoring, and future model improvement
- Feedback is used to improve model performance over time

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2. AI Model Usage and System Architecture

Model Stack Overview

The system uses a **modular AI stack**, where each model has a clearly defined responsibility.

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1. Speech-to-Text Models (Voice Channels)

- Converts voice calls into text
- Optimized for call-center audio
- Outputs time-stamped transcripts

Used only for voice channels and feeds into downstream text-based models.

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2. Intent Classification Model

- Supervised text classification model
- Predicts the customer's primary intent (e.g., refund, outage, billing issue)
- Designed to be fast and interpretable

This model enables routing, prioritization, and analytics.

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3. Sentiment Analysis Model

- Estimates emotional tone (e.g., neutral, frustrated, angry)
- Used as an input signal, not as a decision-maker
- Helps agents respond appropriately

Sentiment is combined with intent, not used in isolation.

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4. Urgency Scoring Model

- Predicts priority level using:
    - Message content

- Sentiment output
- Customer attributes (e.g., premium tier)
- Historical patterns

Outputs a confidence-weighted urgency score rather than a binary label.

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5. Large Language Model (LLM)

The LLM is used **only for assisted reasoning and drafting**, not autonomous decisions.

Primary uses:

- Drafting agent response suggestions
- Summarizing long conversation histories
- Generating structured action recommendations

The LLM never operates without context constraints.

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6. Retrieval-Augmented Generation (RAG)

- Retrieves relevant documents from:
  - Knowledge bases
  - Policies
  - FAQs
  - Past resolved cases
- Injects retrieved content into the LLM prompt

This prevents hallucination and ensures responses stay grounded in enterprise-approved data.

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7. Routing and Decision Engine

- Combines model outputs with business rules
- Handles escalation, assignment, and workflow triggers
- Ensures predictable behavior in high-risk cases

This layer ensures explainability and control.

3. Safety and Accuracy Plan

Human-in-the-Loop Safeguards

- All customer-facing responses require agent approval
- High-risk intents automatically require human review
- Automation is limited to low-risk scenarios

Confidence Thresholds

- Model outputs include confidence scores
- Low-confidence predictions default to human judgment
- Thresholds are configurable per enterprise

Hallucination Prevention

- LLM responses are grounded using RAG
- If relevant documents are unavailable, the system avoids speculative responses
- Agents are alerted when confidence is low

PII Masking and Data Privacy

- Sensitive data is masked before model processing
- No raw PII is sent to language models
- Data retention policies are enforced per enterprise requirements

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Audit Logs and Traceability

- Every model output is logged with:
  - Input
  - Model version

    - Confidence score

- Agent edits and final actions are recorded
- Enables compliance reviews and root-cause analysis

Model Monitoring and Drift Detection

- Continuous monitoring of prediction accuracy
- Alerts for data drift or performance degradation
- Periodic model retraining using validated outcomes

Access Control and Role-Based Permissions

- Agents, managers, and admins have different access levels
- Sensitive insights are restricted based on role
- Prevents misuse of customer data

Closing Note for the AI Blueprint

This AI blueprint prioritizes accuracy, control, and trust over full automation. The system is designed to assist human decision-making, not replace it. By combining deterministic rules with carefully scoped model usage and strong safety mechanisms, the platform delivers practical value while meeting enterprise expectations for reliability and compliance.